together with *data consumers*, start to form connections and compose a network. We call such a network the *mesh*, where the individual *nodes* are data products and consumers.

Even in our small example, we observe a significant operational paradigm change. It encompasses both a shift in the ownership responsibility (from a central data team to the development teams) and the technical challenge of making the new setup work.

Introducing changes in the operational paradigm will result in ripples affecting many areas of your business. To stop these ripples from becoming chaos, we need guiding principles. We briefly introduce them later in this chapter and examine them in detail in chapters 4 to 7. Before that, you must understand our definition of *data mesh* and its nontechnical aspects. Zhamak Dehghani made an incredible effort in curating the idea of the data mesh in 2019 (see "How to Move Beyond a Monolithic Data Lake to a Distributed Data Mesh," http://mng.bz/nNaK). She provided us with all its critical elements and introduced a structured approach to the previously discussed paradigm shift.

Since Dehghani's introduction of the data mesh approach, many data mesh–inspired business-derived and theoretical examples have appeared. A lot of this content might not perfectly fit into the initial description of the data mesh framework. A lot of businesses have seemed somewhat unsure about what exactly conforms to the definition of a data mesh and what doesn't.

In our work, as well as in this book, we opt for solutions that are first and foremost practical (hence the title, *Data Mesh in Action*). Therefore, the data mesh definition we coined aims to be broad and functional and emphasizes *decentralized* efforts to maximize the *value* derived from data.

DEFINITION *Data mesh* is a decentralization paradigm. It decentralizes the ownership of data, the transformation of data into information, and data serving. It aims to increase the value extraction from data by removing bottlenecks in the data value stream. The data mesh paradigm is guided by four principles (described in section 1.4 and in chapters 4 to 7), helping to make data operations efficient at scale: domain ownership, domain data as a product, federated computational governance, and self-serve data platform. Data mesh implementations may differ in scope and in the degree to which they use each principle.